AMBULANCE TRANSMITTER

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <sys/neutrino.h>
#include <sys/netmgr.h>

#define CRITICAL 1
#define ROUTINE 2
#define INTERRUPTED 3

#define HR 1
#define SPO2 2
#define TEMPERATURE 3
#define GPS 4
#define EMERGENCY 5

typedef struct {
   int ambulance_id;
   int parameter;
   int priority;
   int severity;
   double value;
   double latitude;
   double longitude;
} PatientMessage;

int ambulance_id;
int critical_coid;
int routine_coid;
```

```
const char *parameter_name(int parameter) {
    switch (parameter) {
        case HR: return "HEARTBEAT";
        case SPO2: return "SpO2";
        case TEMPERATURE: return "TEMPERATURE";
        case GPS: return "GPS";
        case EMERGENCY: return "EMERGENCY";
        default: return "UNKNOWN";
    }
}

void send_parameter(int parameter, int priority,
                    double value, int severity,
                    double latitude, double longitude) {
    PatientMessage msg;
    int reply = 0;
    int coid;

    memset(&msg, 0, sizeof(msg));

    msg.ambulance_id = ambulance_id;
    msg.parameter = parameter;
    msg.priority = priority;
    msg.value = value;
    msg.severity = severity;
    msg.latitude = latitude;
    msg.longitude = longitude;

    coid = (priority == CRITICAL)
          ? critical_coid : routine_coid;
```

```
    printf("\nSending %s: %s\n",
           parameter_name(parameter),
           priority == CRITICAL ? "CRITICAL" : "ROUTINE");

    if (MsgSend(coid, &msg, sizeof(msg),
                &reply, sizeof(reply)) == -1) {
        perror("MsgSend");
        return;
    }

    if (reply == CRITICAL)
        printf("ACK: Critical alert processed\n");
    else if (reply == ROUTINE)
        printf("ACK: Routine data processed\n");
    else if (reply == INTERRUPTED)
        printf("ACK: Routine processing interrupted\n");
    else
        printf("ACK: Invalid or rejected message\n");
}

int main(void) {

 setvbuf(stdout, NULL, _IONBF, 0);
    int hospital_pid;
    int critical_chid;
    int routine_chid;

    int hr;
    int spo2;
    int severity;
```

```
double temperature;
double latitude;
double longitude;

printf("\nSMART AMBULANCE TRANSMITTER\n");

printf("Enter Ambulance ID: ");
if (scanf("%d", &ambulance_id) != 1)
   return EXIT_FAILURE;

printf("Enter Hospital PID: ");
if (scanf("%d", &hospital_pid) != 1)
   return EXIT_FAILURE;

printf("Enter Critical Channel ID: ");
if (scanf("%d", &critical_chid) != 1)
   return EXIT_FAILURE;

printf("Enter Routine Channel ID: ");
if (scanf("%d", &routine_chid) != 1)
   return EXIT_FAILURE;

critical_coid = ConnectAttach(
   ND_LOCAL_NODE, hospital_pid,
   critical_chid, _NTO_SIDE_CHANNEL, 0);

routine_coid = ConnectAttach(
   ND_LOCAL_NODE, hospital_pid,
   routine_chid, _NTO_SIDE_CHANNEL, 0);
```

```
if (critical_coid == -1 || routine_coid == -1) {
    perror("ConnectAttach");
    return EXIT_FAILURE;
}

while (1) {
    printf("\n============================\n");
    printf("AMBULANCE %d PATIENT DATA\n", ambulance_id);

    printf("Enter Heartbeat (BPM): ");
    if (scanf("%d", &hr) != 1)
        break;

    printf("Enter SpO2 (%%): ");
    if (scanf("%d", &spo2) != 1)
        break;

    printf("Enter Temperature (C): ");
    if (scanf("%lf", &temperature) != 1)
        break;

    printf("Enter Latitude: ");
    if (scanf("%lf", &latitude) != 1)
        break;

    printf("Enter Longitude: ");
    if (scanf("%lf", &longitude) != 1)
        break;

    printf("Enter Emergency Severity (1-5): ");
    if (scanf("%d", &severity) != 1)
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    break;

if (hr < 0 || hr > 300 ||
    spo2 < 0 || spo2 > 100 ||
    temperature < 20 || temperature > 45 ||
    latitude < -90 || latitude > 90 ||
    longitude < -180 || longitude > 180 ||
    severity < 1 || severity > 5) {
    printf("Invalid data. Enter readings again.\n");
    continue;
}

printf("\n--- SENDING CRITICAL PARAMETERS ---\n");

if (severity >= 4) {
    send_parameter(EMERGENCY, CRITICAL,
                severity, severity,
                latitude, longitude);
}

if (hr < 50 || hr > 130) {
    send_parameter(HR, CRITICAL, hr,
                severity, latitude, longitude);
}

if (spo2 < 90) {
    send_parameter(SPO2, CRITICAL, spo2,
                severity, latitude, longitude);
}

if (temperature > 39.0 ||
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   temperature < 35.0) {
   send_parameter(TEMPERATURE, CRITICAL,
                temperature, severity,
                latitude, longitude);
}

printf("\n--- SENDING ROUTINE PARAMETERS ---\n");

if (hr >= 50 && hr <= 130) {
   send_parameter(HR, ROUTINE, hr,
                severity, latitude, longitude);
}

if (spo2 >= 90) {
   send_parameter(SPO2, ROUTINE, spo2,
                severity, latitude, longitude);
}

if (temperature >= 35.0 &&
   temperature <= 39.0) {
   send_parameter(TEMPERATURE, ROUTINE,
                temperature, severity,
                latitude, longitude);
}

send_parameter(GPS, ROUTINE, 0,
            severity, latitude, longitude);

if (severity < 4) {
   send_parameter(EMERGENCY, ROUTINE,
                severity, severity,
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                    latitude, longitude);
        }

        printf("\nAll patient parameters submitted.\n");
        printf("Enter next set of readings.\n");
    }

    ConnectDetach(critical_coid);
    ConnectDetach(routine_coid);

    return EXIT_SUCCESS;
}
```